

Franklin Bien

associate professor

基本信息

姓名: Franklin Bien

所属院校: 蔚山国立科技研究所

地区: South Korea

学科领域

电气工程及其自动化

集成电路设计与集成系统

生物医学工程

微电子科学与工程

电磁场与无线技术

通信工程

研究领域

电力电子与电机驱动

模拟与射频IC设计

低功耗电路与能量管理

混合信号集成技术

生物医学传感与检测技术

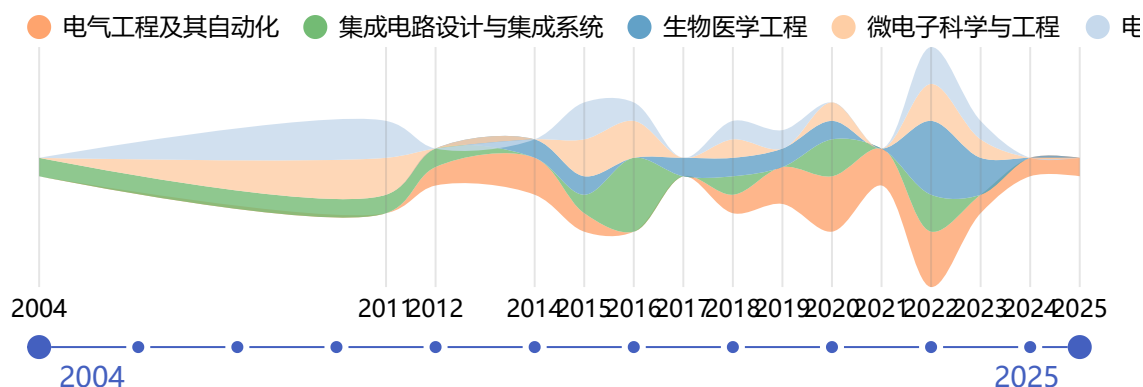
医用生物材料与植入器械

射频集成电路

个人简介

Franklin Bien is an Associate Professor at the School of Electrical and Computer Engineering. His verified email is at unist.ac.kr. His research areas include Analog CMOS IC Design, RF IC Design, Wireless Power Transfer, and Touch Screen Panel Driver IC. His work has been cited 6507 times in total and 4714 times since 2020, with an h - index of 32 (25 since 2020) and an i10 - index of 67 (44 since 2020). He has numerous articles in well - known journals and holds several US patents.

近年发表的论文



Arrangement Free Wireless Power Transfer via Strongly Coupled Electrical Resonances

ADVANCED SCIENCE

Lee, Bonyoung; Kim, Jungho; Jo, Hyunkyeong; Min, Hyunki; Bien, Franklin

被引用次数: 0

2025

A coreless track-type seamless wireless charging system using co-planar wires enabling quasi-free planar movements for mobile logistics robots

APPLIED ENERGY

Jo, Hyunkyeong; Seo, Seoktae; Kim, Jungho; Bien, Franklin

被引用次数: 0

2024

An On-/Off-Time Sensing-Based Load-Adaptive Mode Control of Triple Mode Buck Converter for Implantable Medical Devices

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Park, Woojin; Namgoong, Gyeongho; Choi, Eunho; Bien, Franklin

被引用次数: 1

2023

Real-Time External Compensation System With Error Correction Algorithm for High-Resolution Mobile Displays

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Park, Kyeongmin; Oh, Seunghun; Choi, Dongjin; Shin, Kyeonghan; Cho, Haewan; Bien, Franklin

被引用次数: 1

2023

An ultralow power wearable vital sign sensor using an electromagnetically reactive near field

BIOENGINEERING & TRANSLATIONAL MEDICINE

Seo, Seoktae; Jo, Hyunkyeong; Kim, Jungho; Lee, Bonyoung; Bien, Franklin

被引用次数: 1

2023

A 1.4mW to 119mW, Wide Output Power Range Energy Harvesting System With 2-D Fast MPPT Based on HC for 1k to 50k Illuminated Solar Cell

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Eunho; Namgoong, Gyeongho; Park, Woojin; Kim, Jiwon; Kim, Suhwan; Lee, Bonyoung; Bien, Franklin

被引用次数: 7

2022

Subcutaneously implantable electromagnetic biosensor system for continuous glucose monitoring

SCIENTIFIC REPORTS

Kim, Seongmun; Malik, Jagannath; Seo, Jong Mo; Cho, Young Min; Bien, Franklin

被引用次数: 20

2022

Design of an Out-Folded Patch Antenna With a Zeroth-Order Resonance for Non-Invasive Continuous Glucose Monitoring

IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION

Nguyen, Mai Anh; Bien, Franklin; Byun, Gangil

被引用次数: 9

2022

A 13.56 MHz Wireless Power Transfer System With Fully Integrated PLL-Based Frequency-Regulated Reconfigurable Duty Control for Implantable Medical Devices

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Namgoong, Gyeongho; Park, Woojin; Bien, Franklin

被引用次数: 9

2022

17-aF σ_{rms} Resolution Noise-Immune Fingerprint Scanning Analog Front-End for Under-Glass Mutual-Capacitive Fingerprint Sensors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Park, Kyeongmin; Oh, Seunghun; Heo, Sanghyun; Shin, Sangwoong; Bien, Franklin

被引用次数: 4

2022

A 94% Peak Efficiency Dual Mode Buck Converter With Fully Integrated On-Time-Based Mode Control for Implantable Medical Devices

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Park, Woojin; Namgoong, Gyeongho; Choi, Eunho; Lee, Bonyoung; Park, Hyunjun; Ma, Hyunggun; Bien, Franklin

被引用次数: 7

2022

-11 to 7 dBm Power Range, Triple Band RF Energy Harvesting System With 99.9% Peak Tracking Efficiency and Improved PCE

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Choi, Eunho; Namgoong, Gyeongho; Park, Woojin; Kim, Suhwan; Kim, Jiwon; Lee, Bonyoung; Bien, Franklin

被引用次数: 2

2022

A Real-Time Calibrated Adaptive-ZVS Class-D Power Amplifier for 6.78 MHz Wireless Power Transfer System

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Namgoong, Gyeongho; Park, Hyunjun; Bien, Franklin

被引用次数: 3

2022

Metal Surface Guided-Wireless Power Transfer System for Portable Applications With Multiple Receivers

IEEE ACCESS

Bien, Franklin; Park, Woojin; Lee, Bonyoung; Byun, Gangil

被引用次数: 5

2021

3-12-V Wide Input Range Adaptive Delay Compensated Active Rectifier for 6.78-MHz Loosely Coupled Wireless Power Transfer System

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Namgoong, Gyeongho; Choi, Eunho; Park, Woojin; Lee, Bonyoung; Park, Hyunjun; Ma, Hyunggun; Bien, Franklin

被引用次数: 9

2021

Experimental Realization of Zenneck Type Wave-based Non-Radiative, Non-Coupled Wireless Power Transmission

SCIENTIFIC REPORTS

Oruganti, Sai Kiran; Liu, Feifei; Paul, Dipra; Liu, Jun; Malik, Jagannath; Feng, Ke; Kim, Haksun; Liang, Yuming; Thundat, Thomas; Bien, Franklin

被引用次数: 18

2020

A 32-dB SNR Readout IC With 20-Vpp Tx Using On-Chip DM-TISM in HV BCD Process for Mutual-Capacitive Fingerprint Sensor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Eunho; Kim, Seongmun; Park, Kyoungmin; Heo, Sanghyun; Ma, Hyunggun; Namgoong, Gyeongho; Kim, Haksun; Bien, Franklin

被引用次数: 2

2020

Differential Coded Multiple Signaling Method With Fully Differential Receiver for Mutual Capacitive Fingerprint TSP

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Heo, Sanghyun; Park, Kyeongmin; Choi, Eun-Ho; Bien, Franklin

被引用次数: 12

2020

Bootstrap capacitorless DCM VOT buck converter with dead-time-based off-time calibration for magnetic resonance wireless power transfer

IET POWER ELECTRONICS

Ma, Hyunggun; Namgoong, Gyeongho; Choi, Eunho; Park, Woojin; Lee, Bonyoung; Jang, Heedon; Park, Kyeongmin; Na, Kyungmin; Bien, Franklin

被引用次数: 0

2020

Free Arrangement Wireless Power Transfer System With a Ferrite Transmission Medium and Geometry-Based Performance Improvement

IEEE TRANSACTIONS ON POWER ELECTRONICS

Seo, Seoktae; Jo, Hyunkyeong; Bien, Franklin

被引用次数: 16

2020

Improvement of dual-glucose sensor specificity for prosthetic vascular grafts based on a calibration scheme

IET MICROWAVES ANTENNAS & PROPAGATION

Jang, Heedon; Kim, Seongmun; Ma, Hyunggun; Patel, Ramesh; Yang, Seungboo; Jeong, Jiyun; Seo, Jongmo; Han, Ki Jin; Bien, Franklin

被引用次数: 1

2020

Alignment- and metallic-obstacle-insensitive contactless power transmission system utilizing surface-guided mode

IET MICROWAVES ANTENNAS & PROPAGATION

Song, Seong-Kyu; Malik, Jagannath; Ko, Nak-Young; Park, Woo-Jin; Seo, Seok-Tae; Lee, Bon-Young; Oruganti, Sai Kiran Sharma; Kim, Hak-Sun; Bien, Franklin

被引用次数: 1

2019

Instantaneous Power Consuming Level Shifter for Improving Power Conversion Efficiency of Buck Converter

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ma, Hyunggun; Namgoong, Gyeongho; Choi, Eunho; Bien, Franklin

被引用次数: 7

2019

Step-charging technique for CC/CV mode battery charging with low-cost control components in IPT systems

IET POWER ELECTRONICS

Na, Kyungmin; Ma, Hyunggun; Namgoong, Gyeongho; Kim, Katherine A.; Jung, Jee-Hoon; Bien, Franklin

被引用次数: 6

2018

Differential Difference Amplifier based Parametric Measurement Unit with Digital Calibration

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Ma, Hyunggun; Nam-Goong, Gyeong Ho; Kim, Seulkirom; Lim, Shin-Il; Bien, Franklin

被引用次数: 1

2018

Soft, smart contact lenses with integrations of wireless circuits, glucose sensors, and displays

SCIENCE ADVANCES

Park, Jihun; Kim, Joohee; Kim, So-Yun; Cheong, Woon Hyung; Jang, Jiuk; Park, Young-Geun; Na, Kyungmin; Kim, Yun-Tae; Heo, Jun Hyuk; Lee, Chang Young; Lee, Jung Heon; Bien, Franklin; Park, Jang-Ung

被引用次数: 509

2018

Transparent and flexible fingerprint sensor array with multiplexed detection of tactile pressure and skin temperature

NATURE COMMUNICATIONS

An, Byeong Wan; Heo, Sanghyun; Ji, Sangyoon; Bien, Franklin; Park, Jang-Ung

被引用次数: 356

2018

Electromagnetically induced transparency in sinusoidal modulated ring resonator

APPLIED PHYSICS LETTERS

Malik, Jagannath; Oruganti, Sai Kiran; Song, Seongkyu; Ko, Nak Young; Bien, Franklin

被引用次数: 16

2018

Stretchable Dual-Capacitor Multi-Sensor for Touch-Curvature-Pressure-Strain Sensing

SCIENTIFIC REPORTS

Jin, Hanbyul; Jung, Sungchul; Kim, Junhyung; Heo, Sanghyun; Lim, Jaeik; Park, Wonsang; Chu, Hye Yong; Bien, Franklin; Park, Kibog

被引用次数: 41

2017

Wearable smart sensor systems integrated on soft contact lenses for wireless ocular diagnostics

NATURE COMMUNICATIONS

Kim, Joohee; Kim, Minji; Lee, Mi-Sun; Kim, Kukjoo; Ji, Sangyoon; Kim, Yun-Tae; Park, Jihun; Na, Kyungmin; Bae, Kwi-Hyun; Kim, Hong Kyun; Bien, Franklin; Lee, Chang Young; Park, Jang-Ung

被引用次数: 661

2017

Research on flexible display at Ulsan National Institute of Science and Technology

NPJ FLEXIBLE ELECTRONICS

Park, Jihun; Heo, Sanghyun; Park, Kibog; Song, Myoung Hoon; Kim, Ju-Young; Kyung, Gyouhyung; Ruoff, Rodney Scott; Park, Jang-Ung; Bien, Franklin

被引用次数: 64

2017

A Multi-purpose Fingerprint Readout Circuit Embedding Physiological Signal Detection

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Eom, Won-Jin; Kim, Sung-Woo; Park, Kyeonghwan; Bien, Franklin; Kim, Jae Joon

被引用次数: 3

2016

Optimal inverter logic gate using 10-nm double gate-all-around (DGAA) transistor with asymmetric channel width

AIP ADVANCES

Ryu, Myunghwan; Bien, Franklin; Kim, Youngmin

被引用次数: 5

2016

Graphene-Based Wireless Environmental Gas Sensor on PET Substrate

IEEE SENSORS JOURNAL

Na, Kyungmin; Ma, Hyunggun; Park, Jihun; Yeo, Junho; Park, Jang-Ung; Bien, Franklin

被引用次数: 35

2016

On-Display Transparent Half-Diamond Pattern Capacitive Fingerprint Sensor Compatible With AMOLED Display

IEEE SENSORS JOURNAL

Ma, Hyunggun; Liu, Zhenyi; Heo, Sanghyun; Lee, Jeeyoon; Na, Kyungmin; Jin, Han Byul; Jung, Sungchul; Park, Kibog; Kim, Jae Joon; Bien, Franklin

被引用次数: 37

2016

Wearable, wireless gas sensors using highly stretchable and transparent structures of nanowires and graphene

NANOSCALE

Park, Jihun; Kim, Joohee; Kim, Kukjoo; Kim, So-Yun; Cheong, Woon Hyung; Park, Kyeongmin; Song, Joo Hyeb; Namgoong, GyeongHo; Kim, Jae Joon; Heo, Jaeyeong; Bien, Franklin; Park, Jang-Ung

An Excessive Current Subtraction Technique to Improve Dynamic Range for Touch Screen Panel Applications

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Heo, Sanghyun; Ma, Hyunggun; Bien, Franklin

被引用次数: 1

2016

72 dB SNR, 240 Hz Frame Rate Readout IC With Differential Continuous-Mode Parallel Architecture for Larger Touch- Screen Panel Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Heo, Sanghyun; Ma, Hyunggun; Song, Joohyeob; Park, Kyoungmin; Choi, Eun-Ho; Kim, Jae Joon; Bien, Franklin

被引用次数: 24

2016

A Multilevel Class-D CMOS Power Amplifier for an Out-Phasing Transmitter With a Nonisolated Power Combiner

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Hur, Joonhoi; Kim, Hyungsoo; Lee, Ockgoo; Kim, Kwan-Woo; Bien, Franklin; Lim, Kyutae; Lee, Chang-Ho; Laskar, Joy

被引用次数: 21

2016

Fluorescent light energy harvesting using a capacitive touch screen

ELECTRONICS LETTERS

Kim, Seulkirom; Kim, Katherine A.; Bien, Franklin

被引用次数: 0

2015

77 GHz waveform generator with multiple frequency shift keying modulation for multi-target detection automotive radar applications

ELECTRONICS LETTERS

Quang Nguyen; Park, Myungryeol; Kim, Youngsu; Bien, Franklin

被引用次数: 6

2015

A 200-Mb/s Data Rate 3.1-4.8-GHz IR-UWB All-Digital Pulse Generator With DB-BPSK Modulation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Na, Kyungmin; Jang, Heedon; Ma, Hyunggun; Choi, Yunho; Bien, Franklin

被引用次数: 27

2015

Tracking Optimal Efficiency of Magnetic Resonance Wireless Power Transfer System for Biomedical Capsule Endoscopy

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Na, Kyungmin; Jang, Heedon; Ma, Hyunggun; Bien, Franklin

被引用次数: 110

2015

Highly Transparent and Stretchable Field-Effect Transistor Sensors Using Graphene-Nanowire Hybrid Nanostructures

ADVANCED MATERIALS

Kim, Joohee; Lee, Mi-Sun; Jeon, Sangbin; Kim, Minji; Kim, Sungwon; Kim, Kukjoo; Bien, Franklin; Hong, Sung You; Park, Jang-Ung

被引用次数: 154

2015

A Fully-Differential Correlated Doubling Sampling Readout Circuit for Mutual-capacitance Touch Screens

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kwon, Kihyun; Kim, Sung-Woo; Bien, Franklin; Kim, Jae Joon

被引用次数: 7

2015

Neuro-fuzzy controller for battery equalisation in serially connected lithium battery pack

IET POWER ELECTRONICS

Thi Thu Ngoc Nguyen; Yoo, Hyon-Gi; Oruganti, Sai K.; Bien, Franklin

被引用次数: 22

2015

Wireless Energy Transfer: Touch/Proximity/Hover Sensing for Large Contoured Displays and Industrial Applications

IEEE SENSORS JOURNAL

Oruganti, Sai Kiran; Heo, Sang Hyun; Ma, Hyunggun; Bien, Franklin

被引用次数: 8

2015

An Adaptive Backward Control Battery Equalization System for Serially Connected Lithium-ion Battery Packs

IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY

Ngoc Nguyen; Oruganti, Sai K.; Na, Kyungmin; Bien, Franklin

被引用次数: 66

2014

Tissue-Informative Mechanism for Wearable Non-invasive Continuous Blood Pressure Monitoring

SCIENTIFIC REPORTS

Woo, Sung Hun; Choi, Yun Young; Kim, Dae Jung; Bien, Franklin; Kim, Jae Joon

被引用次数: 33

2014

In-situ Synthesis of Carbon Nanotube-Graphite Electronic Devices and Their Integrations onto Surfaces of Live Plants and Insects

NANO LETTERS

Lee, Kyongsoo; Park, Jihun; Lee, Mi-Sun; Kim, Joohee; Hyun, Byung Gwan; Kang, Dong Jun; Na, Kyungmin; Lee, Chang Young; Bien, Franklin; Park, Jang-Ung

被引用次数: 103

2014

Algorithm for improving SNR using high voltage and differential Manchester code for capacitive touch screen panel

ELECTRONICS LETTERS

Ma, Hyunggun; Heo, Sanghyun; Kim, Jae Joon; Bien, Franklin

被引用次数: 13

2014

Smart In-wheel Generator Using Adaptive DC-DC Converter for Rechargeable Batteries in Electric Vehicles

INTERNATIONAL JOURNAL OF PRECISION ENGINEERING AND MANUFACTURING

Nguyen, Ngoc T.; Ho, Hung V.; Hong, Sung-Tae; Bien, Franklin

被引用次数: 3

2014

A 3.1-4.8-GHz IR- UWB All- Digital Pulse Generator With Variable Channel Selection in 0.13- μ m CMOS Technology

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Yunho; Kim, Youngsu; Hoang, Huy; Bien, Franklin

被引用次数: 17

2012

An Adaptive Technique to Improve Wireless Power Transfer for Consumer Electronics

IEEE TRANSACTIONS ON CONSUMER ELECTRONICS

Hoang, Huy; Lee, Seunggyu; Kim, Youngsu; Choi, Yunho; Bien, Franklin

被引用次数: 133

2012

Low-Power Technique for SRAM-Based On-Chip Arbitrary-Waveform Generator

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Song, Taejoong; Lee, Sang Min; Park, Jongmin; Hur, Joonhoi; Lee, Michael; Kim, Kihong; Lee, Chang-Ho; Bien, Franklin; Lim, Kyutae; Laskar, Joy

被引用次数: 3

2011

A Non-Interruptive Link-Variation Monitoring Circuit for Wireless Sensor Applications

IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS

Kim, Stephen T.; Choi, Jaehyook; Chae, Kwanyeob; Beck, Sungho; Kim, Seong-Hyok; Bien, Franklin; Lee, Chang-Ho; Lim, Kyutae; Laskar, Joy; Tentzeris, Manos M.

被引用次数: 0

2011

An Amplitude and Phase Mismatches Calibration Technique for the LINC Transmitter With Unbalanced Phase Control

IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY

Hur, Joonhoi; Kim, Hyoungsoo; Lee, Ockgoo; Kim, Kwan-Woo; Lim, Kyutae; Bien, Franklin

被引用次数: 20

2011

Fabrication of a Novel Contactless Switch Using Eddy Current Displacement Sensor for Safer Vehicle Brake System

IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY

Kim, Young Su; Choi, Yun Ho; Lee, Jung Min; Noh, Jong Sang; Kim, Jae Joon; Bien, Franklin

被引用次数: 8

2011

Fully integrated UHF RFID mobile reader with power amplifiers using System-in-Package (SiP)

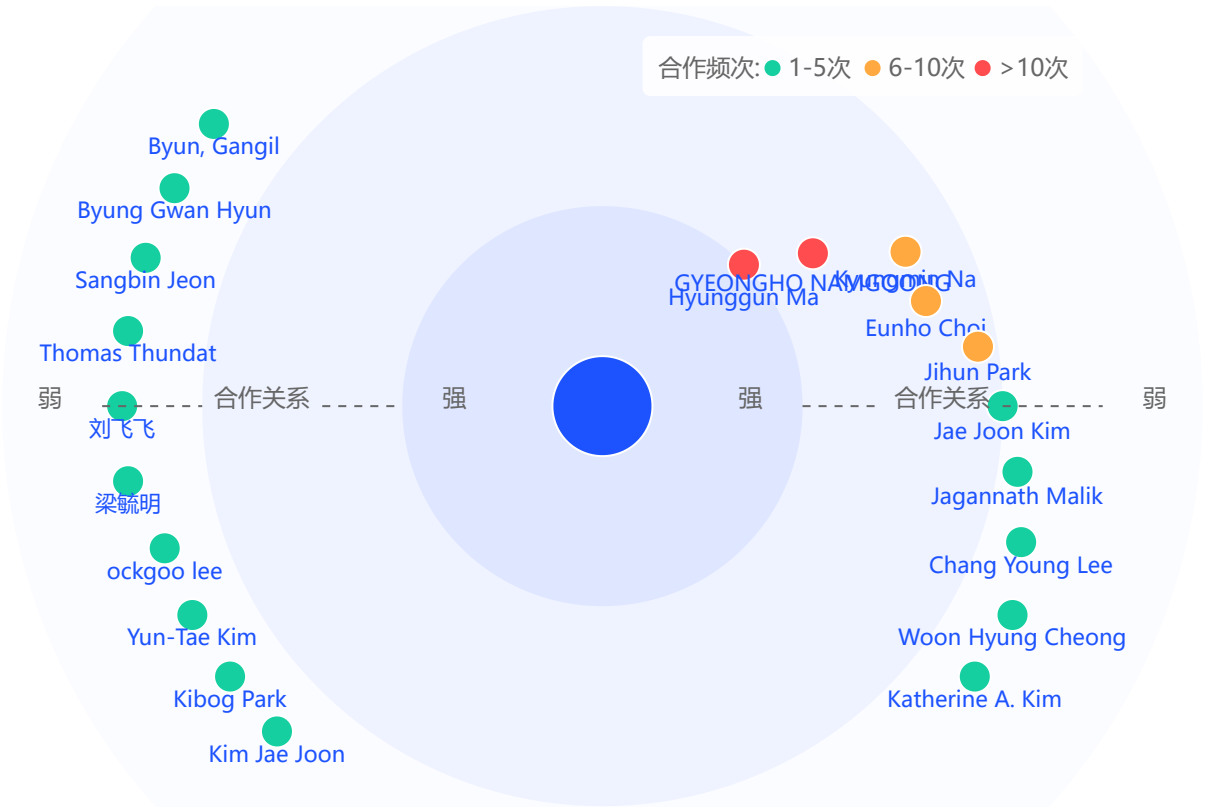
IEICE ELECTRONICS EXPRESS

Yang, Hyemin; Kim, Jongmoon; Bien, Franklin; Lee, Jongsoo

被引用次数: 4

2011

学术合作



Hyunggun Ma	合作次数：15
蔚山国立科技研究所	
GYEONGHO NAMGOONG	合作次数：12
蔚山国立科技研究所	
Kyungmin Na	合作次数：8
蔚山国立科技研究所	
Eunho Choi	合作次数：8
蔚山国立科技研究所	
Jihun Park	合作次数：6
西北大学(埃文斯顿)	

还有 29 个合作者